Routing Protocol Design for Wireless Sensor and IoT Networks: Challenges and Solutions

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1 Introduction

1.1 Background and Motivation

Wireless Sensor Networks (WSNs) and the Internet of Things (IoTs) are at the forefront of two revolutionary technologies that will allow for continuous monitoring and communication over vast areas of various sectors. As explained by Mansour et al. [12] in their review of IoT applications, WSNs have transformed the way a variety of sectors operate, e.g., healthcare, agriculture, smart cities, etc., and environmental monitoring through the use of IoT. The heart of the development of WSNs and IoTs is the need for efficient routing protocol design. Routing protocols form the backbone of data transmission within resource constrained WSN and IoT networks, and the choice of routing protocol can significantly impact the performance of the network and the reliability of the system.

Traditional routing protocols were originally developed for traditional wired or wireless networks. However, the constraints present in sensor networks make it difficult to apply them to this type of environment. The primary constraints of WSNs are the extreme energy limitations of sensor nodes; the highly dynamic nature of the topology of sensor networks; and the necessity to scale the number of nodes in sensor networks to accommodate large numbers of sensor nodes. According to Al-Karaki and Kamal [4], routing protocols for WSNs must be able to accommodate all of these constraints. Foundational surveys by Akyildiz et al. [3] and Akkaya and Younis [2] established the early understanding of these unique challenges. A few examples of routing protocols that are specifically designed to address the needs of WSNs are LEACH (Low-Energy Adaptive Clustering Hierarchy), as described by Heinzelman et al. [8]; and PEGASIS (Power-Efficient Gathering in Sensor Information Systems), introduced by Lindsey and Raghavendra [9].

1.2 Challenges and Scope

Although there have been many advances made in developing routing protocols for WSN-IoT networks, these types of networks still encounter several obstacles today. Examples of

these obstacles include optimizing energy usage; security vulnerabilities; ensuring Quality of Service (QoS); and the ability to adapt to changing network conditions. In addition to identifying the specific challenges associated with routing in low-power networks, Mrabet et al. [13] and Obaidat et al. [14] also provide insight into security vulnerabilities that threaten network integrity, while Shah et al. [16] examine the challenges specific to mobile IoT environments. Rawat and Chauhan [15] further analyze clustering-based approaches that address many of these challenges. This paper discusses the challenges associated with designing routing protocols for WSN-IoT networks, as well as new methods that can improve the performance and sustainability of these types of networks.

1.3 Scope and Organization

The goal of this article is to provide a complete view of the design of routing protocols for wireless sensor networks and Internet of Things (IoT) systems. The article describes some of the main tradeoffs that designers must consider when selecting a routing protocol. The first part of the article focuses on analyzing the principles behind the design and performance of previous routing protocols, rather than just listing and comparing the features of protocols.

Although many researchers (including Heinzelman [8], Lindsey and Raghavendra [9], Manjeshwar and Agrawal [11]) have conducted extensive work on the development of routing protocols for wireless sensor networks and have contributed greatly to our understanding of the field through both their original research and through the development of classification frameworks (protocol overview), much of the current literature has examined how the newly integrated IoT environment has altered the routing landscape. To this end, the article makes use of recent literature (Mansour et al. [12], Shah et al. [16]) that has identified the effects of this new technological environment on the design and implementation of routing protocols.

The article is arranged to provide a comprehensive overview of routing protocol design. The first section provides a conceptual framework for the study of routing protocols and discusses the evolution of wireless sensor networks. The second section will describe the fundamental principles of routing protocols, including: energy-efficient routing mechanisms, classifications of protocols based on their routing methodologies, and considerations of the security of the routing protocols. The third section provides an analysis of different routing strategies and their application in the field. The final section discusses potential ongoing issues associated with routing protocol design, as well as possible future research directions, based on the work of Rawat and Chauhan [15] and Obaidat et al. [14].

2 Background and Conceptual Foundations

2.1 Evolution of Wireless Sensor Networks

From their initial conception in the late 1990s, wireless sensor networks (WSN) have evolved dramatically. The initial concept of wireless sensor networks was to provide surveillance and monitoring to the military. WSN consist of a number of small, usually battery-powered, sensor nodes dispersed throughout an area that gather information about their environment (using sensors) and somehow convey that information to a central location for purposes of analysis.

The early 2000s witnessed a major turning point as researchers started to address the practical issues of Wireless Sensor Networks. Akyildiz et al. [3] presented one of the first comprehensive studies on what makes Wireless Sensor Networks different from traditional wireless networks. Their study identified key differences in Wireless Sensor Networks, stating that nodes are not just small versions of traditional computers connected through wireless media. Rather, they are severely constrained, which changes our thinking about networking. The primary constraint identified was battery life, as replacing batteries for a number of deployed nodes is not feasible.

The technology has advanced significantly, and Wireless Sensor Networks are no longer limited to military surveillance. Environmental monitoring has become one of the key areas where Wireless Sensor Networks are being applied. Scientists can now track various phenomena, ranging from wildlife migrations to volcanic activities. Agricultural monitoring has also become a potential area, where farmers can now track crop conditions. Wireless Sensor Networks are also being applied to the healthcare sector, where patient monitoring has shown promise.

2.2 Fundamental Routing Concepts in WSNs

Routing in wireless sensor networks operates quite differently from routing in traditional networks. In a typical Internet connection, routers have reliable power supplies and computational resources to spare. They can maintain detailed routing tables and perform complex calculations to find optimal paths. Sensor nodes cannot afford such luxury. With limited battery capacity and processing power, every operation must be carefully considered from an energy perspective.

The basic challenge is straightforward: data collected by sensor nodes needs to reach a base station or sink node, but direct transmission over long distances consumes excessive energy. The solution involves multi-hop communication, where data passes through intermediate nodes on its way to the destination. However, this creates new problems. If certain nodes consistently serve as intermediaries, their batteries drain faster than others, potentially creating gaps in network coverage.

Early routing approaches simply tried to minimize the total distance data traveled, assuming shorter paths meant less energy consumption. This turned out to be overly simplistic. The energy cost of transmission does not scale linearly with distance, and more importantly, concentrating traffic through specific nodes creates hotspots that fail prematurely. Al-Karaki and Kamal [4] examined various routing paradigms and found that the optimal approach depends heavily on the specific application and network topology. Akkaya and Younis [2] further classified routing protocols into data-centric, hierarchical, and location-based categories.

Clustering emerged as a particularly effective organizational strategy. Rather than treating all nodes identically, clustering protocols designate certain nodes as cluster heads responsible for aggregating data from nearby sensors before forwarding it toward the base station. This reduces overall transmission costs and provides natural points for data fusion, where redundant information can be eliminated. The challenge lies in selecting cluster heads fairly so that the burden rotates among capable nodes rather than exhausting specific devices.

2.3 Integration of WSNs with IoT

The rise of the Internet of Things brought new dimensions to wireless sensor network research. While traditional WSNs operated as isolated systems tailored to specific monitoring tasks, IoT envisions a world where countless devices communicate seamlessly across different networks and platforms. This shift required rethinking many fundamental assumptions about how sensor networks should be designed and operated.

IoT applications demand more than just data collection. They often require bidirectional communication, where the network not only gathers information but also receives commands to actuate physical processes. A smart building system, for instance, does not just monitor temperature; it adjusts heating and cooling based on occupancy patterns and user preferences. This interactivity introduces latency requirements and quality of service considerations that were less critical in traditional monitoring applications.

Standardization became essential as diverse devices needed to interoperate. Protocol development shifted toward Internet-compatible solutions that could integrate sensor data into broader information systems. This brought advantages in terms of connectivity and data accessibility, but also introduced new security concerns. When sensor networks connect to the Internet, they become potential targets for cyberattacks, requiring authentication mechanisms and encryption that must somehow fit within the tight resource constraints of sensor nodes.

The convergence of WSNs and IoT also changed the scale at which these systems operate. Early sensor networks might involve dozens or hundreds of nodes monitoring a specific location. Modern IoT deployments can encompass thousands of heterogeneous

devices spread across multiple sites, managed through cloud platforms that provide analytics and control interfaces. This expansion makes energy efficiency even more critical, as the cumulative power consumption of massive deployments carries both economic and environmental costs.

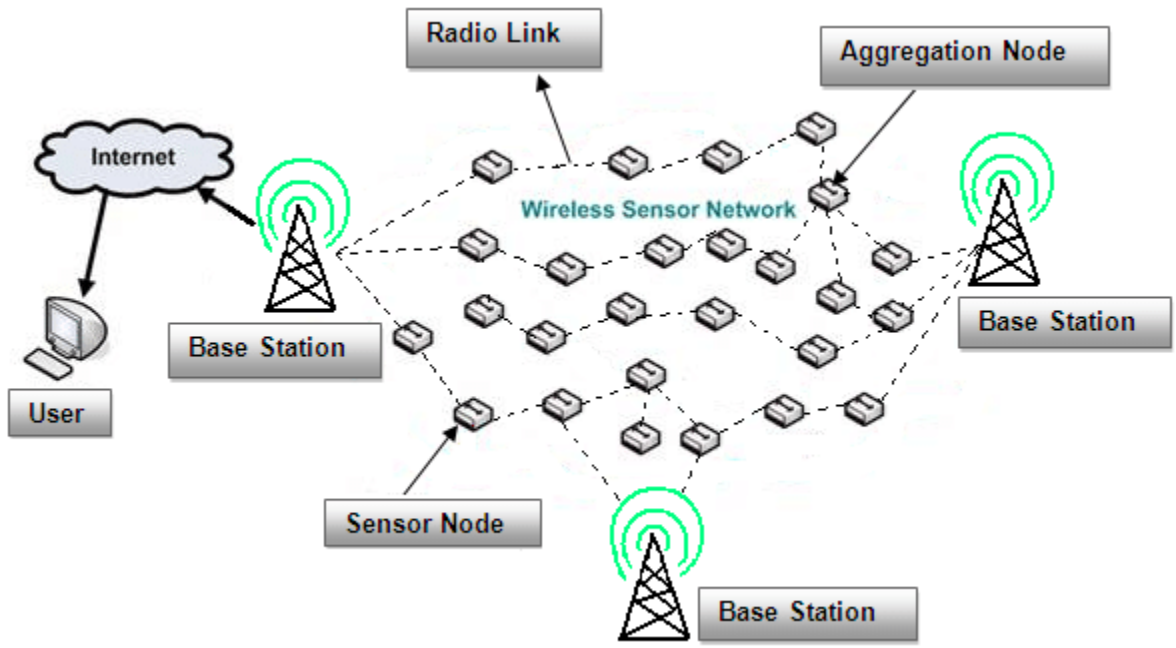

Figure 1: Basic conceptual illustration of wireless sensor network showing sensor nodes, base stations, aggregation nodes, and Internet connectivity based on [3].

Table 1: Comparison of Traditional Networks versus WSN and IoT Networks [3, 12]

Aspect	**Traditional Networks**	**WSN/IoT Networks**
Power Supply	Reliable, continuous power (wired)	Battery-powered, energy-constrained
Node Capabilities	High computational resources, large memory	Limited processing, minimal memory
Network Topology	Relatively static, managed infrastructure	Dynamic, self-organizing, ad-hoc
Primary Objective	Data throughput and bandwidth	Energy efficiency and network lifetime

3 Core Concepts and Approaches

Routing protocol design for wireless sensor networks and Internet of Things environments involves fundamentally rethinking the way we think about communicating with – and through – a network. In contrast with traditional networks where network nodes are generally not constrained by resource availability, sensor networks are severely constrained

in multiple ways and as such impact every design decision. Routing protocol architecture for wireless sensor networks and Internet of Things environments must balance energy efficiency, scalability, and reliability while still being constrained to small hardware.

3.1 Energy-Efficient Routing Mechanisms

Energy efficiency stands as perhaps the most critical consideration in WSN-IoT routing protocol design. Sensor nodes typically run on small batteries that are difficult or impossible to replace once deployed. If a routing protocol causes nodes to deplete their batteries quickly, the entire network fails prematurely, regardless of how well it performs in other aspects. This reality has driven researchers to develop innovative approaches that minimize energy consumption while maintaining network functionality.

The breakthrough came with hierarchical clustering approaches that distribute energy burden more evenly across the network. Heinzelman et al. [8] introduced the Low Energy Adaptive Clustering Hierarchy protocol, commonly known as LEACH, which revolutionized how we think about organizing sensor networks. Rather than having all nodes communicate directly with a distant base station, LEACH groups nearby nodes into clusters. Within each cluster, one node serves as the cluster head, collecting data from member nodes, aggregating it, and forwarding the compressed information to the base station. This approach dramatically reduces the number of long-distance transmissions, which consume far more energy than short-range communication between neighboring nodes.

The elegance of LEACH lies in its rotation mechanism. If the same nodes always served as cluster heads, they would exhaust their batteries quickly while other nodes remained fresh. LEACH addresses this by randomly rotating the cluster head role among all nodes in a cluster. Each node takes its turn bearing the heavier communication burden, ensuring that energy depletion occurs uniformly across the network. Liu et al. [10] further refined these concepts by developing improved cluster head selection algorithms that consider residual energy levels and node locations, creating more balanced cluster distributions that extend network lifetime even further.

Building on these foundational concepts, researchers have explored alternative organizational strategies for energy-efficient data gathering. Lindsey and Raghavendra [9] proposed PEGASIS, which takes a different approach by forming chains of sensor nodes rather than clusters. In PEGASIS, nodes communicate only with their closest neighbors in the chain, and leadership rotates as in LEACH. By eliminating the cluster formation overhead and ensuring every transmission is to a nearby neighbor, PEGASIS achieves even better energy efficiency than LEACH in many scenarios. The trade-off is increased latency, as data must propagate through the entire chain before reaching the base station.

Reactive versus proactive routing represents another fundamental design choice in

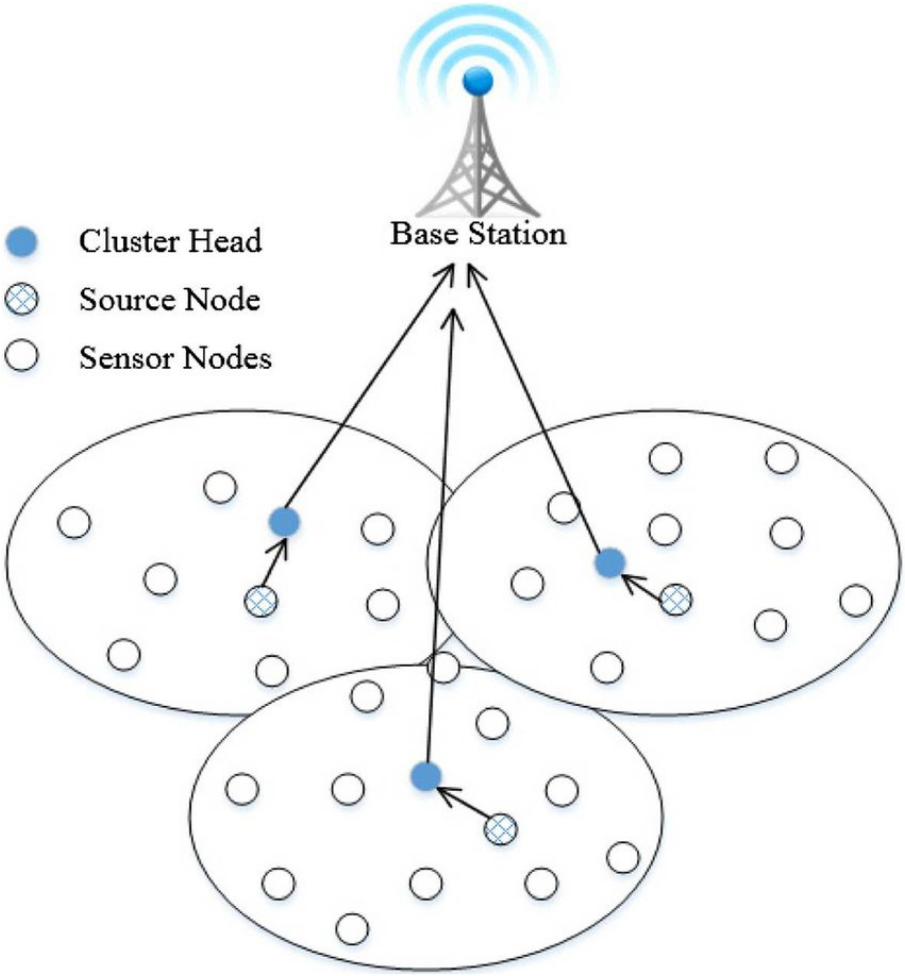

Figure 2: LEACH protocol clustering structure showing cluster heads, source nodes, sensor nodes, and base station communication based on stufy of [8].=

WSN protocols. Proactive protocols maintain routing information continuously, updating paths even when no data needs transmission. Reactive protocols, conversely, establish routes only when data needs transmission. Manjeshwar and Agrawal [11] introduced TEEN, a reactive protocol designed for time-critical applications. TEEN operates based on threshold values: nodes only transmit when sensed parameters exceed critical thresholds or change significantly. For applications like fire detection or intrusion monitoring, this reactive approach makes perfect sense because it avoids wasting energy reporting normal conditions.

3.2 Protocol Classification and IoT Integration

Classification of routing protocols reveals multiple organizational dimensions that affect how networks operate and scale. Al-Karaki and Kamal [4] provide comprehensive analysis of various routing paradigms, examining how different approaches address specific network requirements. Flat routing architectures treat all nodes as equals, each performing

identical functions and making independent decisions. Hierarchical schemes, by contrast, establish different roles for different nodes, with designated coordinators managing subsets of the network. Geographic routing exploits location information, forwarding data based on physical proximity to the destination rather than logical network topology.

Each organizational approach brings distinct advantages and challenges that must be weighed against application requirements. Flat routing offers robustness since no special nodes exist whose failure would disrupt operations, but struggles with scalability as network size grows. Hierarchical approaches scale better and enable efficient data aggregation, but introduce complexity in coordinator selection and create potential single points of failure. The choice between these paradigms depends heavily on deployment scenarios, network size, and the nature of data being collected.

The integration of wireless sensor networks with broader Internet of Things frameworks has introduced new complexities and opportunities. Traditional WSNs operated as isolated systems dedicated to specific monitoring tasks, with custom protocols optimized for particular deployment scenarios. The IoT vision encompasses vastly more ambitious goals: countless heterogeneous devices communicating seamlessly across diverse networks and platforms, all interconnected through Internet protocols. Mansour et al. [12] examine the protocol architectures that enable this integration, noting that successful IoT routing must accommodate devices ranging from tiny environmental sensors to powerful edge computing nodes. The challenge lies in designing protocols flexible enough to work across this spectrum while maintaining efficiency for the most constrained devices.

Mobility introduces yet another layer of complexity to routing design. Static sensor networks, where nodes remain fixed after deployment, can optimize routes based on stable topology. Mobile scenarios, where sensors move or the network structure changes dynamically, require protocols that adapt quickly to topology changes. Shah et al. [16] investigate routing challenges specific to mobile IoT environments, revealing that conventional approaches often fail when nodes move frequently. The fundamental difficulty is that routes established for stationary networks become invalid as nodes relocate, forcing constant route reconstruction that consumes significant energy and reduces data delivery reliability.

3.3 Security and Performance Evaluation

Security considerations have become increasingly critical as sensor networks connect to the Internet and handle sensitive information. Routing protocols must defend against various attack vectors while operating within tight resource constraints that limit the complexity of security mechanisms. Mrabet et al. [13] analyze security threats organized by network layer, demonstrating that each layer of the protocol stack presents unique vulnerabilities that attackers can exploit.

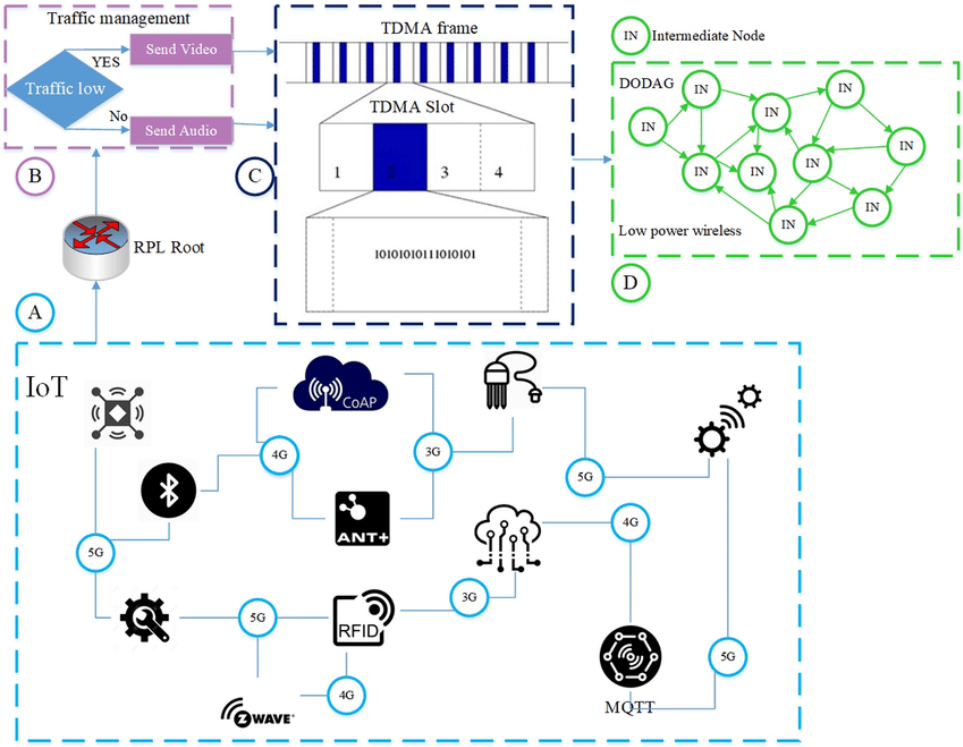

Figure 3: RPL routing protocol architecture for IoT systems showing DODAG structure, traffic management, and heterogeneous device integration adapted and based on [16, 12].

At the routing layer, adversaries might attempt selective forwarding attacks, where compromised nodes drop certain packets while forwarding others to avoid detection. Sinkhole attacks involve malicious nodes advertising attractive routes to lure traffic, then manipulating or discarding the received data. Wormhole attacks tunnel packets across the network through an external channel, disrupting routing topology and potentially enabling other attacks. Defending against these threats requires authentication mechanisms to verify node identities, encryption to protect data confidentiality, and intrusion detection systems to identify anomalous behavior.

Obaidat et al. [14] provide comprehensive examination of security frameworks for IoT, noting that lightweight cryptographic algorithms specifically designed for constrained devices offer promising solutions. The challenge is implementing these security measures on devices with limited computational power and energy budgets. Trust management systems that maintain reputation scores for nodes can help identify compromised devices based on their routing behavior, though these systems themselves consume resources for maintaining and updating trust values.

Performance evaluation of routing protocols requires careful methodology to ensure fair comparison and realistic assessment. Akyildiz et al. [3] emphasize that evaluation must consider the complete system, including hardware characteristics, radio propagation models, and application-specific traffic patterns, rather than relying solely on theoretical analysis or simplified simulations. Network lifetime serves as a primary metric, though its

precise definition varies across studies. Some researchers define lifetime as the time until the first node depletes its battery, while others use the point where network connectivity degrades below a threshold or a percentage of nodes fail.

Additional performance metrics include packet delivery ratio, measuring what fraction of generated data successfully reaches its destination, and end-to-end latency, capturing the time required for data to traverse the network. Control overhead, representing the bandwidth consumed by routing messages rather than application data, directly impacts energy efficiency since every transmitted packet drains battery power. Rawat and Chauhan [15] survey clustering protocols extensively, revealing that the field has produced numerous variants optimizing different performance aspects. Some protocols minimize energy consumption aggressively, accepting longer delays or reduced reliability, while others prioritize data delivery guarantees even at the cost of shorter battery life. This diversity reflects the reality that no single protocol optimally serves all applications.

4 Comparative Discussion

There are many different ways to design routing protocols for wireless sensor networks and the Internet of Things (IoT); each of these designs/choices has its advantages and disadvantages. Understanding the trade-offs involved in choosing between them is important for determining which routing protocol(s) should be used in a given deployment use case. This part discusses the main comparisons among different types of routing protocols, as well as how the design decisions that were made when creating them affect the performance of the network metrics in those networks.

4.1 Hierarchical Clustering versus Chain-Based Routing

The choice of clustering versus chain-based routing will be one of the most significant design decisions for energy-efficient WSN protocols. They both attempt to reduce the total energy used in a WSN by reducing long-range (or long-distance) transmissions. However, they are organized in completely different ways.

An application of the clustering approach can be seen in the LEACH protocol proposed by Heinzelman et al. [8], Group nodes into clusters that coordinate data aggregation at a central base station before sending the combined aggregated results back to the base station. The greatest aspect of this method is the ability to allow all cluster heads to perform this action in parallel with respect to time. This allows for clustering to become ideal for time-sensitive applications; thus, it becomes an attractive solution for many applications that require fast results. Additionally, the ability for cluster head roles to be randomly rotated can allow for the distribution of energy consumption throughout all cluster members; therefore; no one cluster member will end up being severely overburdened by cluster

head activities.

However, there is obviously an extra overhead created by clustering that cannot be overlooked. Clusters require messages to be sent among members to broadcast and establish member roles and perform member elections or may require an additional energy consumption/bandwidth for sending and receiving this necessary message (i.e., sending the necessary message when changing cluster heads every given period of time, etc.). Liu et al. [10] demonstrates that a better method of selecting clusters will greatly extend the useful life of a wireless ad hoc sensor networks, while no algorithm can ultimately remove the essential cost associated with maintaining clusters.

PEGASIS employs chain routing techniques as demonstrated by Lindsey and Raghavendra;[9], Another method they'll be using is for the nodes to avoid cluster formation and rather create a linked chain where each node will only communicate to its closest neighbor for incoming and outgoing information. All nodes will serve as a leader at different intervals and will relay the data to other nodes only after processing it first. There will be no excess transmission overhead as data only transmits to the next node in the chain between neighboring nodes; therefore the energy used during transmission is minimised. The drawback of this system is latency, as data will always report to the base station after being transmitted through every node within the chain, which could increase the amount of time it takes for data to reach the base station in the event of a large network.

Rawat and Chauhan[15] Analysis of tradeoff's across clustering protocols shows that clustering method depends on specific application's need. For example, environmental monitoring systems may be able to afford higher latency by using PEGASIS, as opposed to using LEACH to provide better energy efficiency. However, emergency response systems would require faster data aggregation and delivery using LEACH than they would using PEGASIS. Arafat et al.[6] extend this analysis to highly mobile unmanned aerial vehicle networks, where clustering becomes essential for managing rapid topology changes. Neither approach universally dominates; each excels in its appropriate context.

4.2 Reactive versus Proactive Routing Strategies

In another significant area of comparison, there is the difference between reactive and proactive routing protocols, which differ primarily by when and how they create a way to communicate between nodes. The choice of one or the other type of protocol will significantly affect the amount of energy consumed by the system and its responsiveness to requests.

With proactive protocols, the routing tables for nodes are kept updated continuously so that the routes to destinations can always be ready for use by the application to send its data. As soon as the application wishes to send data, it can be sent immediately without a delay in establishing the route. This is most effective for applications that

exhibit a steady stream of data traffic because the overhead incurred in maintaining the routing tables for these routes is offset by the frequency with which the application uses them. Conversely, if a node occasionally transmits data, then the energy that it uses to keep the routing tables for those routes up to date is essentially wasted.

Manjeshwar and Agrawal [11] TEEN was made to be a protocol that can react quickly in applications where timing is critical because it defines a series of threshold levels and any time a node senses that a parameter is outside of those levels, or has significantly changed from its previous value, it will transition to sending its first message. Because of this, the TEEN protocol is very suitable for applications like fire detection and intrusion detection where there is very little activity most of the time, but when there is abnormal activity, there is a need for a quick response to such uncommon activity. When a node sends its first message of an event, it does take some time to send the message due to the time it takes for the node to establish its routing path; however, TEEN creates an extremely low amount of energy usage while there are no events.

Gnawali et al. [7] The Collection Tree Protocol proposes the use of adaptive beaconing in order to reduce both the latency associated with repairing routes and the overhead associated with control packets. The authors show that hybrid techniques combine the best aspects of reactive and proactive routing methods. Al-Karaki and Kamal [4] analyze the reactive versus proactive choice across various routing paradigms, noting that hybrid approaches maintain routes to frequently contacted destinations while establishing paths to occasional contacts only when needed.

4.3 Security versus Energy Efficiency Trade-offs

The relationship between balancing security needs and energy limitations is one of the tougher comparisons we face today. With the growing number of sensor networks supporting sensitive information that connect to the Internet, the number and types of security mechanisms are essential. Using cryptographic methods and authenticating devices with limited resources come at a significantly higher energy cost which can decrease their overall lifetime.

There is also an increased need for security; hence the relationship between stronger security and more computation and larger message sizes is a direct relationship—everything else being equal, higher-security solutions will result in higher energy consumption than lower-security solutions. [13] The examination of network security threats through different layers by Mrabet et al.has shown that full protection can only be achieved with security mechanisms at many layers. Encrypting the data payload of a message provides confidentiality; however, this will increase the message size and also require encryption operations. Authenticating routing protocols will help protect against the ability to spoof a route; however, each routed packet will contain an authentication tag, and each hop

between two routing nodes will require a verification computation for that authentication tag.

Obaidat et al. [14] Assess various survey security frameworks using algorithmic techniques that promote minimal computation and code size; these lightweight security solutions will significantly mitigate, yet will not eradicate the amount of energy expended on security. Therefore, the question remains how much network lifetime we are willing to forfeit to obtain a sufficient level of security. Medical patient monitoring systems must maintain security with regards to the patient's Health Information, as a data breach will have severe consequences; therefore, while a network monitoring environmental conditions available to the public may still provide sufficient security while maximizing network lifetime.

Almazaideh and Levendovszky [5] Research shows that when it comes to reliable routing protocols, there are trade-offs between reliabilty and energy efficiency. Data delivery becomes more reliable with an acknwoledgment and full-data retransission approach. However, this type of routing results in increased energy consumption and latency. Some applications can tolerate the loss of data from time to time, whereas other require the delivery of each packet with an energy cost being an acceptable trade-off.

When using data aggregation for routing, there is also a higher level of complication added to the security versus efficiency trade off. Abbasian Dehkordi et al. [1] Investigate how different methods to combine multiple sensor measurements before being transmitted can reduce the amount of data sent over the network. These methods have very high energy efficiency, but they also present new security concerns. When nodes that serve as intermediaries aggregate data, they are required to decrypt the message they received, perform the aggregation of those values, and then encrypt the aggregate value again. This requires a significant amount of energy and a sharing of cryptographic keys between aggregation points, thus increasing the availability of a vector for an attacker to exploit.

Table 2 provides a comprehensive comparison of major routing protocols, highlighting their respective strengths, limitations, and suitable application scenarios.

The methods displayed in the table below reflect variable points of time in the design space, with different levels of optimization for Multiple Priorities and trade-offs. LEACH uses a clustering method to optimize energy efficiency while maintaining reasonable latency; however, PEGASIS sacrifices additional latency to achieve a higher level of energy efficiency. TEEN's optimization is primarily targeted at Event-Driven applications where typically very little data needed to be reported and will, therefore, enable TEEN users to expect very low latency. Secure methods will provide a higher level of protection than Secure Duration for Sensitive Data, but each protocol has no advantages across the range of metrics resulting from providing a good match between the protocol used to the requirements of your application.

Table 2: Comparative Analysis of Major WSN-IoT Routing Protocols [8, 9, 11, 5]

Protocol	**Key Strengths**	**Main Limitations**	**Suitable Applications**
LEACH	Distributed cluster formation; randomized cluster head rotation; localized data aggregation; scalable to large networks	Cluster formation overhead; assumes uniform energy distribution; requires synchronization	Periodic environmental monitoring; homogeneous networks with moderate delay tolerance
PEGASIS	Superior energy efficiency through chain organization; minimal transmission distances; no cluster formation overhead	High latency due to sequential data propagation; requires location information; poor mobility support	Long-term environmental monitoring where maximum lifetime is critical and delays are acceptable
TEEN	Extremely low energy consumption during normal periods; immediate response to critical events; threshold-based operation	Route setup delay for first event; requires careful threshold configuration; not suitable for periodic reporting	Time-critical monitoring (fire detection, intrusion alerts, industrial process control)
Secure Routing Protocols	Protection against attacks; data confidentiality and integrity; authentication mechanisms	Increased energy consumption; higher computational overhead; larger message sizes	Medical monitoring; financial transactions; any application handling sensitive data

5 Practical Insights and Use Cases

When designing routing protocols in WSNs and IoT environments, we must go beyond the theoretical area of protocol optimization; although simulations can provide meaningful measurements, real-world challenges typically are underestimated at the point of protocol development. Many real-world factors involved in practical environments (e.g., unpredictable radio interference, hardware limitations, node placement irregularities, energy-depletion imbalance, security threats, and dynamic-topology changes) have significant impacts on how protocols perform when deployed. As such, routing protocols must be designed to optimize not only classical performance metrics (e.g., throughput and delay), but also enable routing protocols to be adaptable, robust, and sustainable over the long term. This section will consider practical issues and representative deployment scenarios where the use of routing strategies is critical to successful operations.

5.1 Energy Optimization and Intelligent Data Handling in Field Deployments

In the design of wireless sensor networks (WSNs) and Internet of Things (IoT) systems, the most important goal is to create energy-efficient systems. Most sensor nodes are constrained in terms of their available resources. The typical limitations for sensor nodes are low battery life and limited processing power. When you look at the types of applications that are typically using sensor nodes (e.g., remote environmental monitoring, remote forest surveillance, and remote structural health monitoring), replacing batteries can either be logistically difficult or economically unfeasible. Because of this, routing protocols must use the least amount of communication overhead possible while ensuring reliable connectivity.

Routing protocols that use clustering to manage the energy imbalance caused by a large number of sensor nodes were one of the earliest practical solutions to this problem. An example of a cluster-based routing protocol is LEACH, which introduced the random rotation of cluster heads to balance energy consumption across sensor nodes.[8]. Equally, the TEEN protocol utilized threshold-based reporting mechanisms for minimizing non-essential data transmissions, providing only when the sensed values went beyond the matrices defined for a given node.[11]. From this evidence of effectiveness, Transmission Scheduling Techniques tend to greatly affect network longevity through established controlled transmission techniques.

Subsequent research enhanced clustering approaches by integrating residual energy awareness, adaptive cluster sizing, and multi-hop cluster-head communication [10, 5].In practice, practical deployments of agriculture have an uneven distribution of soil moisture / temperature sensors over large distances. Cluster formers based on energy consumption ensure that nodes with higher remaining power become leaders within the cluster, thus helping to avoid early network fragmentation. In addition, load balancing techniques reduce excessive forwarding loads on centrally location nodes.

In addition to clustering, strategies for accumulating data that are already within a network (in-network aggregation) have been shown to greatly reduce duplicate or redundant transmissions. To achieve this, various types of aggregation techniques, such as: averaging, compressing data, suppressing duplicate data, and filtering out specific events, all result in lower communications costs and more scalability. [1, 9]. Before transmitting data wirelessly, it is helpful to perform some intermediate processing in order to save energy. Smart irrigation systems can often get by using aggregated summaries of what has happened at each field, rather than streaming raw data all thetime.

Aggregation, though, presents some challenges. For xample, if too much aggregation takes place, it may create issues with data granularity and could affect how accurate the results arewhen trying to detect anomalies. For this reason, the routing protocol needs

to adpt the amount of aggregation used dynamically based on application needs, the amount of time alowed for the data to be transmitted, andthe amount of congestion with the network.[12]. The need to provide immediate notification of abnormal sensor readings requires that IoT devices used in health monitoring systems skip the aggregation layer while returning to every other action in the system. However, normal sensor readings will continue through the aggregated pathway.

In addition, real-world case studies highlight the need for reliable link quality estimating tools and adaptive transmission power controlling techniques. Many environmental influences affect the reliability of a link, including humidity, physical obstruction, and electromagnetic interference. Employing proper routing protocols that use link quality indicators and dynamically adjust power settings can improve the successful packet delivery ratio and also reduce unnecessary retransmissions. [3, 4]. This adaptability becmes particularly important in industrial IoT settingswhere metallic structures and machineri frequently disrupt signal propagations.

5.2 Mobility, Security, and Large-Scale Operational Considerations

Modern IoT infrastructures increasingly incorporate mobility into their design. Applications using mobility include vehicular networks, UAV-supported collection of data, wearable sensors, and disaster recovery systems. Most of these applications rely on dynamic topologies and cannot use traditional routing systems, as they change frequently. Therefore, a solution to this issue is to use mobility-determined routing protocols along with delay-tolerant communication protocols to resolve problems with intermittent connectivity and rapid route failures.[16, 6].

UAV-Assisted Sensor Networks Use Drones to Collect Buffered Data from Ground Nodes. This Reduces the Need for Multi Hop Forwarding to Continuously Send Data to a Remote Location. By Collecting and Storing Data, Static Nodes Consume Less Energy and Provide Coverage in Geographically Sparse Areas. Thus, Synchronization and Scheduling of Data Transfers is Important to Prevent Data Buffers from Overfilling during Times When Drones are Not Present.

Security is Another Area of Concern for Real-World Deployments. For Example, WSN and IoT Routing Have Many Security Vulnerabilities such as Sinkhole Attack (the Ability of One Node to Block Other Nodes and Create an Alternate Route), Wormhole Attack (a Node in One Location Can Pretend to be at Another Location When Creating Routes), Selective Forwarding (Sending Only a Small Amount of Data When the Total Amount Should Be for Many Nodes), and Node Impersonation (Pretending to Be Another Node). These threats Will Impact the Security and Integrity of Data and the Reliability of the Network.[13, 14]. To address networking security threats, practical routing protocols

have built-in methods for lightweight authentication, encryption, and trust evaluation. The downside to this is the added computational and communication overhead of implementing security features which may cause energy efficiency reduction. Thus, there is a requirement for designers to weigh the resources available with the strength of the security features implemented.

Continuously growing from a prototype to a city-wide scale presents unique challenges especially when considering network scalability. Hierarchical routing architectures, location-based routing protocols, and load-balancing mechanisms assist in providing continually improved performance as the number of nodes increase.[2, 15].As congestion and the routing overhead increase due to a lack of scalable structuring, they reduce throughput and increase latency.

For example, using smart city traffic monitoring systems, an incident report that is latency-sensitive would be given priority over standard telemetry updates and be forwarded sooner than other non-urgent data using quality-of-service differentiation (QOS) in the routing protocol. Additionally, using adaptive prioritization mechanisms, such as QOS, improve the overall reliability of an incident report, and enhance the user experience of all data that is delivered.

In addition to inter-operability with heterogeneous devices, the type of device used can affect routing decisions practically. For example, when deploying an IoT, multiple device types, such as Wi-Fi to backhaul to wired/ wireless, may work together. A routing protocol must support multiple communication standards, as well as develop optimized strategies to route across multiple layers.[12]. Failure to address heterogeneity can lead to bottlenecks at gateway nodes and inefficient resource utilization.

Overall, successful routing protocol design in WSN and IoT systems requires a holistic perspective that integrates energy awareness, adaptive mobility support, scalable structuring, and security reinforcement. Practical deployments consistently demonstrate that robustness and adaptability are as important as theoretical optimality. Protocols that perform efficiently under controlled simulations may encounter significant degradation in real environments unless designed with operational variability in mind.

5.3 Large-Scale Deployment and Heterogeneity Challenges

There are many challenges associated with the scalability of routing protocols as deployment moves from small, experimental designs to larger city-wide infrastructures which are often vastly underestimated during the initial design phase of the project. When networks increase in number from a few hundred to tens of thousands of nodes, the associated routing overhead and the infrastructure management complexity exponentially increase as well. In addition, using a proper hierarchical architecture can help maintain performance but must be configured properly in order to prevent bottleneck situations

from occurring [2, 15].

A primary difficulty of deploying large-scale systems involves addressing the varying characteristics (heterogeneities) of devices and their interactions. IoT systems today encompass many types of heterogeneous devices: low-power Sensors, Wi-Fi Gateways, Cellular Backhaul Link equipment, and Cloud-infrastructure components. Each one of these devices operates on its own set of communication standards/protocols along with having different energy consumption characteristics and different levels of processing capacity. The routing protocols must be able to develop a level of support for communicating between those various/heterogeneous technologies while providing optimized performance for each device class [12].

It is important to create a quality of service differentiation when dealing with varying data types that have different priority levels. Emergency alerts must be delivered immediately and with very little latency compared to regular environmental readings, which typically can wait longer and could follow a low-energy path or route [16]. Typically, gateway nodes act as bottlenecks in heterogeneous networks (i.e., networks made up of various communications technologies) by aggregating traffic from many sensor nodes and bridging between them.

Several real-world examples show how theoretical optimizations don't equal practical gains in the real-world environment of actually using devices built using protocols. Examples include variations in the environment, imperfections in hardware, limitations of firmware/software (the computer program that controls the operation of a device), etc. Protocols must also include measures to adjust to these factors so that they can continue to perform effectively even when facing challenges due to environmental conditions, hardware failures, etc [3, 2].

6 Challenges and Open Issues

Despite significant advancements in routing protocol design for Wireless Sensor Networks (WSNs) and Internet of Things (IoT) systems, numerous challenges remain unresolved. While early routing protocols primarily focused on energy efficiency, modern IoT environments introduce additional complexities such as heterogeneity, large-scale deployment, mobility, and security vulnerabilities. Practical implementations often reveal limitations that are not fully addressed in theoretical models. This section discusses major unresolved challenges and open research directions that continue to motivate further investigation.

6.1 Energy Constraints, Scalability, and Heterogeneity

While energy-efficient routing methods (e.g., clustering, duty cycling, energy-aware) have extended the lifetime of a WSN [8, 11], maintaining balanced energy usage among all

nodes becomes a challenge especially in dynamic environments [10, 5]. Due to the variance in the amount of data flowing through nodes and changes to the networks topology, some nodes (e.g., nodes located close to the sink) will run out of energy sooner than other nodes. If not managed correctly, this leads to the partitions in the network and reduced connectivity [2, 4].

As network size grows from tens of thousands or even a million nodes, routing design becomes more complex due to scalability issues with regards to large networks [3]. Protocols that can be implemented effectively in small scale test environments perform poorly in larger scale testing due to excessive control overhead in larger implementations. Hierarchical routing and clustering provide some solutions to these problems [15], but the additional overhead incurred from maintaining the clusters and the increased complexity of synchronizing clustered nodes is significantly greater when the density of the network increases. In addition, there are continuing challenges associated with managing routing table size, limiting discovery to flooding-based methods, and mitigating the effects of broadcast storms.

IoT is becoming really complex because of all the different devices being used together in a way they have never been used before to create IoT applications [12]. Devices being used in modern IoT systems include computers with different capabilities for computing, communicating, and consuming power (i.e., devices). One example of this would be that sensor nodes that communicate wirelessly using low power IEEE 802.15.4 need to communicate with other types of devices, like WiFi gateways, in order to access the cloud. Therefore, routing protocols have to be able to support multiple-tiered and cross-layer routing so that all heterogeneous devices used in an IoT application can be integrated into one routing framework to allow for seamless integration acros all heterogeneous devices used in an IoT application is an open research question.

Another major challenge will be finding the right balance between energy efficiency and Quality of Service (QoS) when using IoT devices. There are many applications for which low latency and high reliability are critical (such as healthcare monitoring and industrial automation) that could conflict with energy efficiency. A routing algorithm needs to be developed that can dynamically weigh the QoS requirements vs. energy efficiency as a future concern [16].

6.2 Security Vulnerabilities and Trust Management

Safety is an essential subject within routing networks used for wireless sensor networks (WSN) and Internet of Things (IOT). Routing attacks pose greater risks to our devices (which have limited capabilities) because they target our devices' routing capabilities. There are several types of routing attacks: Sinkhole attacks, Wormhole attacks, Sybil attacks, and Selective Forwarding attacks.[13, 14].System reliability can be adversely

affected due to attacks against routing paths by manipulating them, injecting false information and breaking communication flows.

Lightweight cryptographic methods and authentication schemes have been developed, however adding them to routing protocols creates additional computational and communication loads.[13]. Security improvements often lead to shortened life spans of the network because they require more processing power; at the same time, no routing protocol has been developed that provides sufficient security guarantees without substantially increasing the amount of energy they consume.

A new type of routing protocol called a "trust-based routing framework" has been proposed to help address this situation. In a trust-based framework, each node will evaluate the behavior of its neighbors before forwarding packets. However, trust evaluation mechanisms are vulnerable to false reporting and collusion-based attacks. In addition, trust metrics need to be maintained, which involves additional memory and communication resources. The scalability of trust-based routing in large deployments of Internet of Things (IoT) is still an open questio

An additional concern that is emerging relates to privacy, particularly with respect to smart cities and health care applications. While the contents of data may be encrypted, the traffic patterns associated with them may still disclose information about users' activities or the layout of their physical infrastructure. [14]. The protection of meta-data privacy preservation for maintaining good routing efficiency continues to be an unsolved area of research.

6.3 Mobility, Reliability, and Real-World Deployment Limitations

Mobility creates changing topologies that violate the basic assumption of traditional routing protocol designs. In vehicular IoT applications such as drone-assisted sensing and wearable networks the routing protocols utilized will have to support rapidly changing routes.[16, 6]. Frequent link breakages contribute to packet loss as well as to increased routing overhead when transmission takes place in highly populated or high-speed environments.

Although delay-tolerant networking techniques can alleviate the effects of temporary disconnections by allowing nodes to store, or 'buffer', packets for some period of time, buffer management, synchronization and latency constraints remain unresolved issues. Excessive buffering will not be acceptable for time-sensitive applications such as emergency responders.

The stability of links is impacted by the unpredictable nature of the environment. Things like physical objects, inclement weather, and electromagnetic radiation can greatly affect the stability of a link.[3]. Most routing protocols make the assumption that links

will remain stable over time; however, this is not true in most instances. Finding a solution that will allow us to implement real-time link quality estimates, as well as the use of adaptive transmission control, while minimizing their contribution to overall system overhead is still an open problem.

At last, the distinction between simulation and real-life application continues to be an ongoing problem. The majority of routing protocols are tested in a perfect simulation environment with homogenous node arrangement and uncomplicated energy models. [2].In real-world usage, the performance of a deployed system can be limited by many factors such as hardware capability, firmware, and environmental differences; as a result, there will need to be established benchmark standards or frameworks for testing routing protocols in the future.

While notable advancements in the area of routing protocols exist for wireless sensor networks and Internet of Things (IoT) interfaces, unfulfilled projects addressing energy balancing, scalability, security, mobility adaption, and practical world robustness remain subject to continued investigation. Addressing these open problems is essential to build large capacity, secure IoT systems that will remain intact through time.

7 Future Directions

Research is still underway in both Routing Protocols for Wireless Sensor Networks (WSNs) and Internet of Things (IoT) systems. As new devices become available, as applications require more from them, and as those devices are deployed more widely, routing protocols will see continued development. Future work will also increasingly focus on adaptive intelligence at the network edge as well as stronger and more efficient security and privacy guarantees for these constrained devices. In addition to these areas, there will be an emphasis on standard testbeds and interoperability frameworks that can bridge the multiple technologies involved. The subsections below describe some of the most promising directions combining recent trends with traditional open issues and will include references to relevant literature.

7.1 AI-driven and Edge-native Adaptive Routing

The use of on-device and edge-intelligent hardware for intelligent routing decisions is seen as a major area of growth. Machine learning and reinforcement learning techniques will be used to help routers (and any nodes with few resources) make predictions about link characteristics (like quality), predict congestion, and dynamically adjust their routing metrics instead of just following a pre-defined heuristic. [16, 12].By enabling direct adaptation of the application without needing to communicate with the cloud, edge-based inference allows for route optimization and localized anomaly handling near real time.

[15].Compact models and light-weight training and updating methods that meet the energy and memory requirements of the user's environment are necessary for practical use. The use of federated and split-learning methods to keep raw sensor data local and only share model updates may be particularly useful in protecting users' privacy and improving routing performance.[13, 5]. Hybrid strategies that combine traditional protocol fallbacks with model-driven rules provide a greater level of robustness for routing systems when there is uncertainty regarding the learned model, or when the environment is subject to drift.

7.2 Security, Privacy, and Trustworthy Routing for Constrained Devices

As Wireless Sensor Networks (WSNs) and Internet of Things (IoT) technologies play increasingly crucial roles, routing layers must be designed with stronger security and privacy safeguards while remaining sensitive to resource budget constraints. Therefore, future routing protocols must include lightweight cryptographic primitives, flexible and device lifecycle-aware certificate and key management, and route-aware intrusion detection to flag suspicious behavior at the earliest possible stage of forwarding activity.[14, 13]. In relation to urban sensing and healthcare applications, privacy-preserving routing, such as methods to conceal traffic patterns or apply selective perturbation of metadata while maintaining utility, will be an essential tool.[12, 15]. While both trust metrics and reputation systems are still viable, they require more robust defenses against collusion and dishonest reporting. An example of an innovative approach to provide accurate trust calculations with limited communication overhead is to combine cross-layer signals with temporal behavior models.[2].

Co-designing routing and security will ensure that protection is built into the routing decisions being made, rather than as an afterthought.

7.3 Standardization, Interoperability, and Real-World Evaluation

The community needs to cooperate on developing interoperable systems and establishing established evaluation protocols for transitioning innovative technologies from laboratories into widespread use. Ongoing improvements in low-power adaptation layers for IPv6, and routing standards have made it easier to implement these solutions; however, the differences in the various radio technologies, gateways, and application backend systems make it very difficult to achieve smooth operation across the entire system. [3, 2]. There are measures being taken to standardize message formats, metric semantics, and management interfaces, and this will reduce the amount of frication in integration. In

addition, it is also important to create reproducible real-world benchmarks and share test beds that provide us with variability and real-time mobile characteristics in addition to any interference. The results obtained from simulations that only simulate an ideal environment will not always generalize.[9, 15]. By using measurements of methodology and using artifacts and open datasets among routing solutions, progress can be made more quickly and comparisons to routing alternatives can occur in a fair manner.

8 Conclusion

8.1 Summary of Key Findings

This article highlighted key issues with both the routing protocol design of wireless sensor networks (WSNs) and the Internet of Things (IoT) and their possible solutions. The primary limiting factor in the design of WSN routing protocols remains energy consumption and the desire for energy efficiency guides the decisions made regarding the design of these routing protocols. There are trade-offs between latency and energy consumption associated with hierarchical clustering approaches (e.g., LEACH) and chain-based approaches (e.g., PEGASIS). Reactive routing protocols, such as TEEN, are designed to work well in event-driven applications while proactive routing protocols have advantages when WSNs are used for continuous monitoring type applications.

Security and reliability have become more important as WSNs are integrated with larger Internet-based infrastructures. The trade-off between the deployment of secure communication mechanisms and energy efficiency will continue to challenge the design of wireless sensor routing protocols. Data aggregation techniques can provide significant energy savings, but will complicate the implementation of end-to-end security for data collected via a WSN. The testing methodology for performance evaluation needs to dive deeper than performing testing exclusively in a controlled environment via hardware and environmental conditions that occur on the actual environment (e.g., physical attributes of the node placement and/or environmental disturbances). The analysis of the major protocols indicates no protocol will provide the optimum solution for all deployment scenarios.

8.2 Practical Implications for Protocol Selection

Careful matching of protocol characteristics and application requirements is required for the successful deployment of both WSN and IoT routing protocols. For example, environmental monitoring systems can accept higher latencies for increased network lifetimes, therefore energy-optimized protocols are an appropriate fit. In contrast, time-critical applications such as industrial control systems or emergency response require low-latency

delivery of data, even at a premium in energy usage.

The heterogeneity of modern IoT deployments requires protocols to operate seamlessly between different types of devices and communication technologies. Gateway bottlenecks and interoperability issues are common in large heterogeneous networks. QoS differentiation allows prioritization of critical data while routing routine telemetry through energy-efficient paths. Mobility introduces dynamic changes in topology when using mobile applications, challenging traditional routing assumptions, and requiring adaptive protocols capable of quickly reconfiguring routes.

Applications across multiple domains have different levels of security requirements. For example, medical monitoring and financial applications cannot compromise on the security of data due to energy costs. However, public environmental monitoring has a reasonable ability to sacrifice some level of security strength for the sake of network lifetime.

8.3 Future Outlook and Closing Remarks

It is important to create a quality of service differentiation when dealing with varying data types that have different priority levels. Emergency alerts must be delivered immediately and with very little latency compared to regular environmental readings, which typically can wait longer and could follow a low-energy path or route. Typically, gateway nodes act as bottlenecks in heterogeneous networks (i.e., networks made up of various communications technologies) by aggregating traffic from many sensor nodes and bridging between them.

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